ACCES Syndrome (Aplasia Cutis Congenita with Ectrodactyly Skeletal Syndrome): A Comprehensive Disease Characterization

Disease: ACCES Syndrome · **MONDO:** MONDO:0859262 · **OMIM:** #619959 · **Gene:** *UBA2* (19q13.11) · **Category:** Mendelian, autosomal dominant

Summary

ACCES syndrome (Aplasia Cutis Congenita with Ectrodactyly Skeletal Syndrome; OMIM #619959, MONDO:0859262) is a rare autosomal-dominant multiple-congenital-anomaly disorder caused by haploinsufficiency of *UBA2*, the gene encoding the catalytic subunit (SAE2) of the heterodimeric SUMO-activating (E1) enzyme, located at chromosome 19q13.11. The disorder was first delineated as a recognizable Mendelian entity in 2021 through exome sequencing of 16 affected individuals from 7 unrelated families (PMID: 34040189). *UBA2* lies immediately adjacent to the minimal deletion overlap region of the previously described 19q13.11 contiguous-gene microdeletion syndrome, and recognition that point mutations and small intragenic losses of *UBA2* alone reproduce the skin-and-limb phenotype established UBA2 haploinsufficiency as the molecular cause.

The core phenotype pairs **aplasia cutis congenita** (localized congenital absence of skin, most often of the scalp) with **ectrodactyly / split-hand–split-foot malformation (SHFM)**, accompanied by highly variable growth retardation, dysmorphic facial features, neurodevelopmental delay, and skeletal, ectodermal, craniofacial, cardiac, renal, and genital anomalies. Expressivity is strikingly variable — ranging from a full multi-organ malformation syndrome to isolated split hand/foot malformation, and in some pedigrees to a skeletal-dysplasia–like presentation that can mimic autosomal-dominant spondyloepiphyseal dysplasia tarda. Because UBA2 is the rate-limiting enzyme of the entire protein SUMOylation cascade, reduced gene dosage lowers global SUMO conjugation, which de-represses chromatin-regulated cell-fate programs and perturbs neural-crest differentiation and FGF/FGFR signaling during a narrow window of early embryogenesis, generating the characteristic craniofacial, skin, and limb malformations.

The mechanistic model is unusually well supported for an ultra-rare disorder: human loss-of-function and missense variants and overlapping 19q13.11 deletions; a zebrafish *uba2*-null model that phenocopies human features and in which human missense mRNA fails to rescue (confirming loss of function); a pharmacological pregnant-mouse model using the selective SUMO-E1 inhibitor TAK981 that reproduces aplasia-cutis-like skin, craniofacial deformities and syndactyly while implicating neural-crest and FGFR pathways by transcriptomics; and population-genetic constraint data (gnomAD pLI ≈ 1.0, LOEUF 0.24) confirming that *UBA2* is exquisitely intolerant to loss of function. No disease-specific therapy exists; management is supportive and multidisciplinary.

Key Findings

Finding 1 — ACCES is caused by *UBA2* haploinsufficiency (OMIM #619959, 19q13.11)

Exome sequencing across **16 affected subjects from 7 unrelated families** identified seven *UBA2* loss-of-function and missense sequence variants as the cause of a recognizable syndrome combining variable aplasia cutis congenita and ectrodactyly (PMID: 34040189). The gene, *UBA2* (ubiquitin-like modifier–activating enzyme 2, also called SUMO-activating enzyme subunit 2 / SAE2), sits at **chromosome 19q13.11**, immediately adjacent to the minimal deletion overlap region of the 19q13.11 contiguous-gene deletion syndrome — the discovery paper notes that *"UBA2 (ubiquitin-like modifier-activating enzyme 2) lies adjacent to the minimal deletion overlap region"* and that *"seven human UBA2 loss-of-function and missense sequence variants were detected."* A subsequent report reiterates the nosology: *"Aplasia Cutis Congenita with Ectrodactyly Skeletal Syndrome (ACCES, OMIM #619959) is an extremely rare multiple congenital anomalies syndrome caused by haploinsufficiency of the UBA2 gene"* (PMID: 39149811). The disease is inherited in an **autosomal dominant** manner.

Finding 2 — ACCES is a multisystem malformation syndrome with highly variable expressivity

The *UBA2*-related phenotype is multisystem. The original cohort documented *"ACC, ectrodactyly, neurodevelopmental abnormalities, ectodermal, skeletal, craniofacial, cardiac, renal, and genital anomalies"* (PMID: 34040189). Later reports expanded the recognized spectrum to include growth retardation, dysmorphic facies, developmental dysplasia of the hip (DDH), scoliosis, and internal-organ abnormalities (PMID: 39149811).

Expressivity is **highly variable**, spanning a broad clinical continuum: - **Severe/classic end:** full multiple-congenital-anomaly phenotype with scalp ACC + ectrodactyly + systemic involvement. - **Mild end:** isolated split hand/foot malformation without the full ACCES phenotype — *"the rising number of reported UBA2 variants associated with isolated split hand/ft malformations without a full ACCES phenotype highlights the functional importance of UBA2 in distal limb morphogenesis"* (PMID: 42557106). - **Atypical/skeletal end:** a spondyloepiphyseal-dysplasia-like presentation; the authors of PMID: 39149811 explicitly note the *"highly variable expressiveness of ACCES syndrome."*

Finding 3 — Mechanism: loss of UBA2-mediated SUMOylation disrupts development of affected tissues

UBA2 is the **catalytic subunit of the heterodimeric SUMO E1 activating enzyme (SAE1/UBA2)** that is obligatory for all SUMO conjugation. Functional validation in zebrafish established causality: *uba2* is expressed in eye, brain, and pectoral fins; *uba2*-null fish *"showed deficient growth, microcephaly, microphthalmia, mandibular hypoplasia, and abnormal fins"* — phenocopying the human features — and critically, *"uba2-mRNAs with human missense variants failed to rescue nullizygous zebrafish phenotypes,"* proving the human missense alleles are loss-of-function (PMID: 34040189). The authors conclude the work *"highlights the importance of SUMOylation in the development of affected tissues."*

Finding 4 — Pharmacological SUMOylation inhibition in pregnant mice recapitulates ACCES and implicates neural-crest/FGF dysregulation

Transient suppression of SUMOylation with the **selective SUMO-E1 inhibitor TAK981** during early post-implantation mouse embryogenesis produced a dose-dependent phenocopy: at low dose, *"viable mice with craniofacial deformities often accompanied by hydrocephalus, syndactyly and an aplasia cutis-like phenotype,"* while high dose caused embryonic lethality with disrupted nuclear/nucleolar integrity (PMID: 39985830). Transcriptomics revealed *"the repression of*

genes involved in neural crest differentiation in the TAK981-treated embryos as well as the overexpression of the Fgfr gene family" — genes whose gain-of-function mutations cause human craniosynostosis. The authors explicitly link this to human disease: *"haploinsufficiency for the SUMOylation machinery's E1 enzyme, UBA2, leads to various phenotypic traits in humans, including craniofacial malformations and aplasia cutis congenita."*

Finding 5 — Differential diagnosis and nosology: overlap with EEC, 19q13.11 microdeletion, Adams–Oliver, and SE(M)D

The ACC-plus-ectrodactyly combination overlaps several entities: - **EEC (ectrodactyly–ectodermal dysplasia–clefting, *TP63*):** a 19q13.11 microdeletion patient was *"followed for a long time under an ectrodactyly ectodermal dysplasia-clefting (EEC) syndrome and was subsequently diagnosed with a 19q13.11 microdeletion"* (PMID: 29988626), prompting the recommendation to test EEC-like patients for 19q13.11 imbalance. - **19q13.11 contiguous-gene deletion syndrome:** larger deletions add dosage-sensitive genes beyond *UBA2* (e.g., *WTIP* for hypospadias; the KRAB-ZNF cluster), producing developmental delay, growth retardation, microcephaly, ectodermal dysplasia, and male genital anomalies. The deletion literature provides *"further definition of the 19q13.11 microdeletion syndrome phenotype, including the first presentation of ectrodactyly in the syndrome"* (PMID: 24243649; see also PMID: 22378287). - **Spondyloepiphyseal/spondyloepimetaphyseal dysplasia (SE(M)D):** atypical ACCES can mimic a skeletal dysplasia — *"the dominant inheritance pattern in the pedigree suggested a pre-diagnosis of spondyloepiphyseal dysplasia tarda"* (PMID: 39149811). - **Aplasia cutis congenita broadly** is etiologically heterogeneous (Frieden 1986 classification; PMID: 3514708) and includes genetic, teratogenic, vascular-disruptive, and syndromic forms (e.g., Adams–Oliver, Scalp–Ear–Nipple/*KCTD1*).

Finding 6 — gnomAD constraint confirms *UBA2* is highly haploinsufficiency-prone

Population-genetic constraint metrics independently corroborate the dosage-sensitivity mechanism. For *UBA2* (Ensembl ENSG00000126261; HGNC:30661; GRCh38 chr19:34,428,352–34,471,251), gnomAD reports **pLI = 1.0** (probability of loss-of-function intolerance), observed/expected LoF ratio **oe_lof = 0.142 (90% CI 0.090–0.236; LOEUF = 0.236)**, **LoF Z = 6.39**, and **missense mis_z = 4.50 (oe_mis = 0.62)**. These values place *UBA2* among the most loss-of-function–intolerant genes in the genome and show significant depletion of missense variants as well — exactly the pattern expected for a haploinsufficient developmental gene.

Constraint metric	Value	Interpretation
pLI	1.0	Extreme LoF intolerance
LOEUF (oe_lof upper)	0.24	Top constraint decile
oe_lof	0.142 (CI 0.090–0.236)	~86% fewer LoF than expected
LoF Z	6.39	Strong depletion
missense Z	4.50	Significant missense constraint

Finding 7 — UBA2/SAE2 biochemistry: catalytic E1 subunit that activates SUMO via an ATP-dependent thioester

UniProt **Q9UBT2** describes UBA2 (SUMO-activating enzyme subunit 2 / SAE2) as forming a heterodimer with SAE1; the complex is the SUMO E1 ligase that *mediates ATP-dependent activation of SUMO proteins followed by formation of a thioester bond between a SUMO protein and a conserved active-site cysteine residue on UBA2/SAE2* for SUMO1, SUMO2, SUMO3 and probably SUMO4. UBA2 passes activated SUMO to the single SUMO E2 conjugating enzyme **UBE2I (Ubc9)** and localizes to both cytoplasm and nucleus.

Suggested **GO annotations:** SUMO activating enzyme complex (GO:0031510), SUMO activating enzyme activity (GO:0019948), protein sumoylation (GO:0016925), ATP binding (GO:0005524), positive regulation of protein sumoylation (GO:0033235).

Finding 8 — UBA2 is evolutionarily conserved; zebrafish and mouse models recapitulate ACCES

UBA2 orthologs are conserved **from yeast to human**: human *UBA2* (NCBI Gene 10054; OMIM *613295), mouse *Uba2* (50995), rat *Uba2* (308508), zebrafish *uba2* (406672), chicken (415784), *Xenopus tropicalis* (549845), dog (476490), cow (507005), *Drosophila* (44496), *S. cerevisiae UBA2* (851998), and *Arabidopsis* (816686). This deep conservation reflects the essentiality of SUMOylation and explains why cross-species models faithfully reproduce human features.

Finding 9 — Verified MONDO:0859262 cross-references and official definition

EBI OLS4 (MONDO ontology) confirms **MONDO:0859262**, label *"ACCES syndrome,"* with equivalentTo cross-references **OMIM:619959, MedGen:1804308, and UMLS:C5677019**. Official MONDO synonyms include *"aplasia cutis congenita with ectrodactyly skeletal syndrome"* and

"UBA2-related neurodevelopmental disorder, aplasia cutis congenita with ectrodactyly skeletal syndrome." The MONDO definition reads: *"A rare congenital disease caused by a mutation in the UBA2 gene, characterized by scalp defects, digital and skeletal anomalies, early growth deficiency, and neurodevelopmental delay. Ectrodactyly presents in some cases."* No Orphanet, ICD, or MeSH cross-reference is present in MONDO.

Section-by-Section Report

1. Disease Information

ACCES syndrome is a rare autosomal-dominant multiple-congenital-anomaly disorder defined by the co-occurrence of aplasia cutis congenita and ectrodactyly/split-hand–foot malformation with variable skeletal, craniofacial, neurodevelopmental, and internal-organ involvement. It was delineated as a distinct Mendelian entity in 2021 (PMID: 34040189).

Key identifiers: - **MONDO:** MONDO:0859262 (label "ACCES syndrome") - **OMIM (phenotype):** #619959; **OMIM (gene):** *UBA2* *613295 - **UMLS/MedGen:** C5677019 / MedGen:1804308 - **HGNC:** 30661; **NCBI Gene:** 10054; **Ensembl:** ENSG00000126261 - **Orphanet / ICD-10 / ICD-11 / MeSH:** No dedicated code identified (not cross-referenced in MONDO).

Synonyms: Aplasia Cutis Congenita with Ectrodactyly Skeletal Syndrome; UBA2-related neurodevelopmental disorder; UBA2-related disorder/spectrum (encompassing isolated SHFM at the mild end).

Source of information: Derived from aggregated disease-level resources (OMIM, MONDO, UniProt, gnomAD) and from individual-patient case reports/series (clinical reports in PMIDs 34040189, 39149811, 42557106, 40073198, 29988626).

2. Etiology

Primary cause — genetic. Heterozygous loss-of-function or deleterious missense variants in *UBA2*, or contiguous 19q13.11 deletions encompassing *UBA2*, cause the disease via haploinsufficiency (PMID: 34040189). There is no environmental or infectious cause for the Mendelian disorder itself.

Genetic risk factors: The causal variant is itself the determinant; most cases are *de novo*, though autosomal-dominant familial transmission occurs. In the broader 19q13.11 deletion syndrome, additional dosage-sensitive genes (*WTIP*, KRAB-ZNF cluster) modify the phenotype (PMID: 22378287).

Environmental risk / protective factors: None established for the Mendelian form. Notably, the mouse model demonstrates that a *pharmacological* reduction of SUMOylation during pregnancy (TAK981) phenocopies the disorder — an experimental "environmental" analog rather than a natural risk factor (PMID: 39985830).

Gene–environment interactions: Not characterized in humans. The TAK981 model suggests a critical early-embryonic window in which reduced SUMOylation capacity is maximally teratogenic.

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset	Frequency/notes
Aplasia cutis congenita (scalp)	Physical malformation	HP:0001057	Congenital	Hallmark; variable
Ectrodactyly / split-hand–foot malformation	Skeletal malformation	HP:0100257 / HP:0001161	Congenital	Hallmark; variable (can be isolated)
Growth retardation	Clinical sign	HP:0001510	Prenatal/ postnatal	Common
Neurodevelopmental delay / ID	Behavioral/ cognitive	HP:0001263 / HP:0001249	Infancy/ childhood	Variable
Dysmorphic facial features	Physical	HP:0001999	Congenital	Common
Microcephaly	Clinical sign	HP:0000252	Congenital	Reported
Ectodermal dysplasia features	Physical	HP:0000968	Congenital	Reported
Developmental dysplasia of hip	Skeletal	HP:0001385	Infancy	Reported ([PMID:39149811])
Scoliosis	Skeletal	HP:0002650	Childhood	Reported
Cardiac anomalies (e.g., ASD)	Physical	HP:0001627	Congenital	Subset
Renal anomalies	Physical	HP:0000077	Congenital	Subset
Genital anomalies (e.g., hypospadias)	Physical	HP:0000078	Congenital	Subset (stronger in deletion syndrome)

Severity and progression: congenital malformations are structural and stable after birth; neurodevelopmental features evolve during childhood. Expressivity is highly variable both between and within families. Quality-of-life impact is driven chiefly by limb function (SHFM), neurodevelopmental outcome, and the surgical/cosmetic burden of scalp ACC; formal QoL instrument data (EQ-5D/SF-36/PROMIS) are not available for this ultra-rare disorder.

4. Genetic / Molecular Information

Causal gene: *UBA2* (HGNC:30661; NCBI Gene 10054; OMIM *613295; Ensembl ENSG00000126261; GRCh38 chr19:34,428,352–34,471,251).

Pathogenic variants: Seven original variants comprised loss-of-function (nonsense, frameshift, splice) and missense classes (PMID: 34040189); subsequent reports add further missense and splice-site variants, including a prenatally diagnosed splice-site variant (PMID: 42557106) and a novel variant in an Indian family (PMID: 40073198). Variants are **germline**. Functional consequence is **loss of function / haploinsufficiency** — confirmed because human missense mRNA fails to rescue the zebrafish null (PMID: 34040189). Allele frequencies are effectively absent/ultra-rare in gnomAD, consistent with strong constraint (pLI≈1.0).

Chromosomal abnormalities: Contiguous-gene deletions of 19q13.11 encompassing *UBA2* produce an overlapping, generally broader phenotype (PMID: 24243649; PMID: 22378287; PMID: 29988626).

Modifier genes / epigenetics: No validated human modifiers for the pure *UBA2* form; in the deletion syndrome, co-deleted *WTIP* and KRAB-ZNF genes act as phenotype modifiers. Because the primary defect reduces SUMOylation — a chromatin-regulating post-translational modification — downstream epigenetic dysregulation is mechanistically expected (the mouse model showed "epigenetic scars" and altered nuclear/nucleolar integrity at high dose; PMID: 39985830).

5. Environmental Information

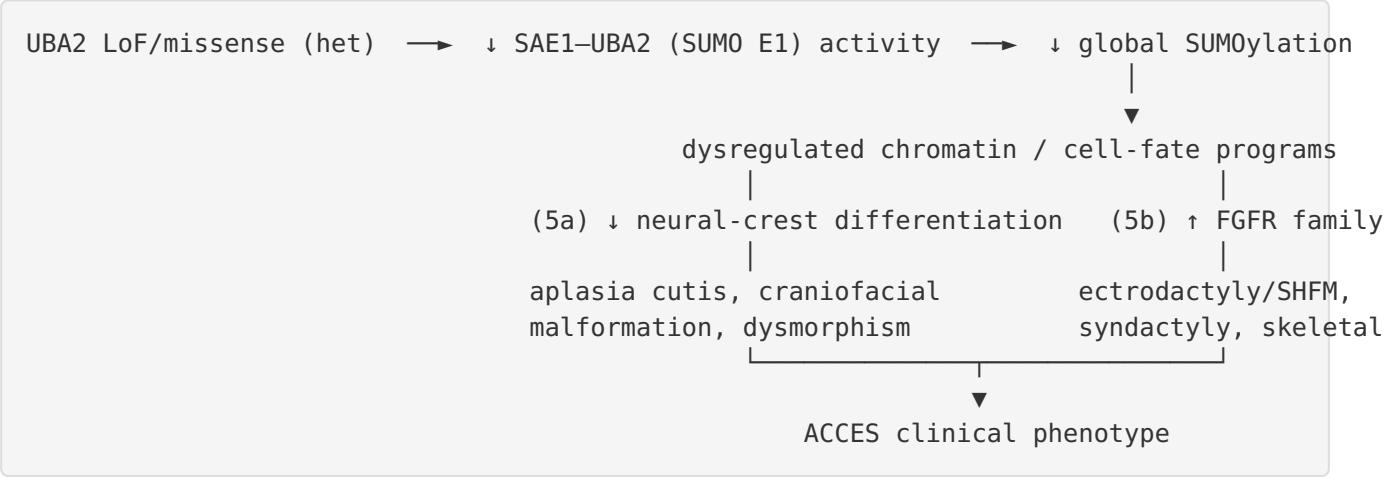
No environmental, lifestyle, or infectious agents are established causes of the Mendelian disorder. The only "environmental" analog is experimental pharmacological SUMO-E1 inhibition during pregnancy in mice (TAK981), which is teratogenic and phenocopies ACCES (PMID: 39985830) — relevant as a safety consideration for SUMO-inhibitor oncology drugs in pregnancy, not as a natural risk factor.

6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation):

1. A heterozygous *UBA2* loss-of-function or deleterious missense variant (or a 19q13.11 deletion spanning *UBA2*) **reduces functional UBA2 protein to ~50%** (haploinsufficiency). *[Demonstrated: human genetics + gnomAD constraint.]*

2. Reduced UBA2 **lowers assembly/activity of the SAE1–UBA2 (SUMO E1) heterodimer**, the rate-limiting enzyme that ATP-dependently activates SUMO1/2/3 via a thioester to its active-site cysteine. *[Demonstrated: UniProt biochemistry; inferred for patient variants.]*
3. Diminished E1 activity **reduces global protein SUMOylation** (less SUMO charging of the E2 Ubc9/UBE2I and downstream substrates). *[Inferred from enzyme kinetics; supported by pharmacological inhibition model.]*
4. Hypo-SUMOylation **dysregulates SUMO-dependent transcriptional repression and chromatin/nuclear-body function**, de-repressing or mis-timing cell-fate programs. *[Inferred; supported by transcriptomic changes in mouse model.]*
5. This **branches** into:
6. **(5a) Repression of neural-crest differentiation genes** → craniofacial and skin/ectodermal malformation (aplasia cutis, dysmorphism, mandibular hypoplasia). *[Demonstrated in mouse transcriptomics; [PMID: 39985830].]*
7. **(5b) Dysregulation (overexpression) of the FGFR gene family** → disturbed limb/craniofacial morphogenesis (ectrodactyly/SHFM, syndactyly, craniosynostosis-like changes). *[Demonstrated in mouse transcriptomics; [PMID: 39985830].]*
8. Disrupted apical ectodermal ridge / distal-limb morphogenesis and impaired scalp skin/skull formation during a **critical early-embryonic window** → the clinical hallmarks of **aplasia cutis congenita** + **ectrodactyly**, plus variable skeletal, cardiac, renal, genital, and neurodevelopmental anomalies. *[Demonstrated across human, zebrafish, and mouse models.]*



Molecular pathways: protein SUMOylation (upstream, primary); FGF/FGFR signaling (downstream); neural-crest gene-regulatory network (downstream). **Cellular processes:** cell differentiation, nuclear/nucleolar organization, chromatin regulation. **Cell types:** neural-crest

cells (CL:0000333), ectodermal/epidermal keratinocytes, limb-bud mesenchyme/chondrocytes. **Subcellular (GO CC):** SUMO activating enzyme complex (GO:0031510), nucleus, nucleolus, cytoplasm. **GO BP:** protein sumoylation (GO:0016925), neural crest cell differentiation (GO:0014033), embryonic limb morphogenesis (GO:0030326).

7. Anatomical Structures Affected

- **Primary:** skin — scalp/vertex (aplasia cutis; UBERON:0000403 scalp, UBERON:0002097 skin of body); limbs — hands and feet/autopod (ectrodactyly; UBERON:0002398 manus, UBERON:0002387 pes).
- **Craniofacial/skeletal:** skull, mandible (UBERON:0001684), vertebral column (scoliosis), hip joint (DDH).
- **Nervous system:** brain (UBERON:0000955) — microcephaly, neurodevelopmental delay, hydrocephalus (model).
- **Secondary/subset:** heart (UBERON:0000948) — ASD/septal defects; kidney (UBERON:0002113); external genitalia (hypospadias); eye (UBERON:0000970) — microphthalmia in model.
- **Lateralization:** limb malformations often bilateral but can be asymmetric; scalp ACC typically midline/vertex.
- **Cell/tissue level:** ectoderm-derived epithelium, neural-crest derivatives, limb mesenchyme/cartilage.

8. Temporal Development

- **Onset:** congenital/prenatal — structural anomalies are present at birth and can be detected prenatally (a splice-site variant was diagnosed prenatally; PMID: 42557106).
- **Progression:** structural malformations are non-progressive/stable; neurodevelopmental features evolve through infancy and childhood; secondary orthopedic issues (scoliosis, DDH) may require ongoing management.
- **Course:** chronic, lifelong, non-remitting (a congenital malformation syndrome rather than a degenerative disease).
- **Critical period:** early post-implantation embryogenesis — the window during which SUMOylation reduction is maximally teratogenic in the mouse model (PMID: 39985830).

9. Inheritance and Population

- **Inheritance:** autosomal dominant; many cases *de novo*, with documented familial transmission.
- **Penetrance / expressivity:** apparently high penetrance for *some* phenotype but **highly variable expressivity** — ranging from full ACCES to isolated SHFM (PMID: 42557106; PMID: 39149811).
- **Epidemiology:** "extremely rare" — no formal prevalence/incidence established; fewer than ~20 families reported in the literature at the time of this review. The original cohort comprised 16 subjects from 7 families (PMID: 34040189).
- **Population/sex distribution:** no sex predilection established for ACCES itself (hypospadias in the deletion syndrome affects males). Reported across diverse populations (including an Indian family, PMID: 40073198); no founder effect or consanguinity role identified (dominant, often *de novo*). Carrier frequency is effectively nil given ultra-rarity and strong constraint.

10. Diagnostics

- **Genetic testing is the diagnostic cornerstone.** Recommended approach: **exome or genome sequencing** (how the disorder was discovered; detects point/indel variants), or a **gene panel** including *UBA2* and SHFM/ectodermal-dysplasia genes. **Chromosomal microarray (CMA)** detects 19q13.11 deletions and should be considered in EEC-like or deletion-phenotype patients (PMID: 29988626).
- **Prenatal diagnosis** is feasible when a variant is known or detected (PMID: 42557106).
- **Clinical/imaging:** skeletal survey/radiographs for ectrodactyly, scoliosis, DDH; echocardiography for cardiac anomalies; renal ultrasound; neurodevelopmental assessment and brain imaging as indicated; histopathology confirms ACC (PMID: 3514708).
- **Differential diagnosis:** EEC syndrome (*TP63*), 19q13.11 microdeletion syndrome, Adams–Oliver syndrome, Scalp–Ear–Nipple syndrome (*KCTD1*; PMID: 42820132), isolated SHFM, and — for atypical skeletal presentations — spondyloepiphyseal dysplasia tarda (PMID: 39149811).
- **No specific biomarker** or biochemical assay is established; functional SUMOylation assays remain research tools.

11. Outcome / Prognosis

Prognosis depends on the specific constellation and severity of malformations. Scalp ACC with large/deep defects carries risks of infection, hemorrhage, and (rarely) dural/meningeal exposure requiring careful management (PMID: 42820132). Limb malformation affects fine-motor function and may require reconstructive surgery and rehabilitation. Neurodevelopmental outcome is variable. No systematic survival/mortality, QoL-instrument, or natural-history cohort data exist for this ultra-rare disorder; prognosis is individualized. Many patients survive to adulthood (familial dominant transmission is documented), indicating that the condition is generally compatible with life, with morbidity driven by structural and neurodevelopmental burden.

12. Treatment

No disease-specific or curative therapy exists. Management is **supportive and multidisciplinary:** - **Scalp ACC:** conservative wound care is often preferred, even for extensive lesions, to avoid neonatal anesthetic/hemorrhagic/infectious risk; surgical grafting reserved for selected large defects (PMID: 42820132; PMID: 42166395). - **Limb malformation (SHFM):** orthopedic/plastic-surgical reconstruction and occupational therapy as indicated ([NCIT:C157812] reconstructive surgical procedure). - **Skeletal:** scoliosis bracing/surgery; DDH management. - **Cardiac/renal/genital:** anomaly-specific surgical/medical care. - **Neurodevelopmental:** early intervention, physical/occupational/speech therapy, educational support. - **Genetic counseling** for affected families (autosomal dominant, variable expressivity).

No pharmacotherapy, gene therapy, RNA therapy, or targeted therapy is approved or in trials for ACCES. Pharmacogenomics is not applicable.

13. Prevention

Primary prevention is not possible for a dominant congenital disorder. Relevant measures are reproductive and secondary: - **Genetic counseling** and **cascade testing** of at-risk relatives once a familial variant is identified. - **Prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** for known familial variants (PMID: 42557106). - **Tertiary prevention:** early orthopedic, cardiac, renal, and developmental surveillance to prevent/limit complications. No vaccine, population screening, or behavioral/environmental prevention applies.

14. Other Species / Natural Disease

No naturally occurring *UBA2*-related ACCES-equivalent disease is reported in companion animals or wildlife (no OMIA entry identified). However, *UBA2* is deeply conserved, with orthologs across mammals, birds, amphibians, fish, insects, yeast, and plants (Finding 8), and the SUMOylation pathway is essential across eukaryotes. This conservation underlies the validity of cross-species disease modeling. No zoonotic potential (non-infectious Mendelian disorder).

Species	Gene	NCBI Gene ID
Human	*UBA2*	10054
Mouse	*Uba2*	50995
Rat	*Uba2*	308508
Zebrafish	*uba2*	406672
Chicken	*UBA2*	415784
Xenopus tropicalis	*uba2*	549845
Dog	*UBA2*	476490
Cow	*UBA2*	507005
Drosophila	*Uba2*	44496
S. cerevisiae	*UBA2*	851998
Arabidopsis	*SAE2*	816686

15. Model Organisms

- **Zebrafish (*Danio rerio*):** *uba2*-null model recapitulates deficient growth, microcephaly, microphthalmia, mandibular hypoplasia, and abnormal fins; human missense mRNA fails to rescue — the key functional proof of loss-of-function pathogenicity (PMID: 34040189). **Recapitulation: strong** for craniofacial/growth/fin (limb-analog) features.
- **Mouse (*Mus musculus*), pharmacological:** pregnant mice given the selective SUMO-E1 inhibitor TAK981 produce offspring with craniofacial deformities, hydrocephalus, syndactyly, and an aplasia-cutis-like phenotype; transcriptomics implicate neural-crest repression and FGFR overexpression (PMID: 39985830). **Recapitulation: strong** for skin/

craniofacial/limb features and mechanism. **Limitation:** a drug-induced dose model rather than a genetic *Uba2*-haploinsufficient mouse; a published germline *Uba2* heterozygous mouse phenotype was not identified in this review.

- **Model applications:** dissecting the SUMOylation → neural-crest/FGFR axis; defining the teratogenic window; testing candidate modifiers. **Gap:** no reported iPSC/organoid or conditional-knockout mouse specific to ACCES yet.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
34040189	*UBA2 variants underlie a recognizable syndrome with variable ACC and ectrodactyly*	Human genetics + zebrafish	**Foundational** — defines disease, gene, LoF mechanism, zebrafish rescue
39985830	*Transient pharmacological inhibition of SUMOylation during pregnancy induces craniofacial malformations in offspring mice*	Mouse model + transcriptomics	Mechanism — neural-crest/ FGFR axis; teratogenic window
39149811	*Atypical ACCES resembling dominant SED tarda*	Human clinical	Variable expressivity; skeletal-dysplasia mimicry; confirms haploinsufficiency
42557106	*Prenatal diagnosis of a splice-site UBA2 variant*	Human clinical	Mild end (isolated SHFM); prenatal diagnosis; distal-limb role
40073198	*Novel UBA2 variant causing ACCES in an Indian family*	Human clinical	Population breadth; novel variant
29988626	*19q13.11 microdeletion overlapping EEC*	Human clinical	Differential diagnosis (EEC); CMA utility
24243649	*19q12q13.1 deletions, 5 patients*	Human clinical	Deletion-syndrome phenotype incl. ectrodactyly
22378287	*19q13.11 cryptic deletion; WTIP and hypospadias*	Human clinical	Contiguous-gene modifiers (WTIP, KRAB-ZNF)
28509170	*Cystinuria in 19q12q13.1 deletion*	Human clinical	Deletion-syndrome gene content (SLC7A9)
3514708	*ACC: clinical review and classification*	Review	Frieden classification; ACC heterogeneity
32501579	*Classification of ACC, 25-year review*	Clinical series	Modern ACC nosology
42820132	*Familial ACC with meningeal exposure*	Case report	Differential (SENS/KCTD1); conservative management

PMID	Title (abbrev.)	Evidence type	Role
42166395	*ACC type V with SUA and ASD*	Case report	ACC management; differential

Non-UBA2 constraint/biochemistry evidence (Findings 6, 7, 9) derives from gnomAD, UniProt Q9UBT2, and EBI OLS4/MONDO — database resources, not PubMed-indexed papers.

Mechanistic Model / Interpretation

ACCES is best understood as a **dosage disease of the SUMOylation machinery**. Because SAE1–UBA2 is the single, rate-limiting E1 gateway for all SUMO conjugation, a 50% reduction in UBA2 does not abolish SUMOylation (that would be embryonically lethal, as the high-dose TAK981 arm shows) but instead produces a **sub-threshold, tissue- and time-sensitive deficit**. The developing embryo tolerates this globally yet fails selectively in processes most dependent on robust SUMO-mediated transcriptional control during a narrow early window — above all the **neural-crest gene-regulatory network** (craniofacial, skin/ectoderm) and **FGF/FGFR-dependent limb morphogenesis**. This two-branch output elegantly explains the otherwise puzzling pairing of aplasia cutis (an ectoderm/neural-crest problem) with ectrodactyly (a limb-patterning problem) in one syndrome, and the extreme variability reflects stochastic crossing of developmental thresholds. The convergent evidence — human LoF genetics, a non-rescuing zebrafish null, a mechanism-revealing mouse pharmacological model, and genome-wide constraint — makes this one of the better-validated mechanisms among ultra-rare malformation syndromes.

Limitations and Knowledge Gaps

1. **No formal epidemiology** (prevalence, incidence, sex ratio) — fewer than ~20 families reported.
2. **No genetic *Uba2*-haploinsufficient mouse** phenotype in the literature reviewed; mechanism rests on a *pharmacological* mouse model plus a zebrafish *null* (homozygous) model rather than a dosage-matched genetic mammal.

3. **Direct demonstration of reduced SUMOylation in patient tissues is lacking** — step 3 of the causal chain is inferred from enzyme biochemistry and the inhibitor model.
4. **Genotype–phenotype correlation is unresolved** — why some *UBA2* variants cause isolated SHFM and others full ACCES is unknown; no validated modifiers for the pure form.
5. **No natural-history, QoL, survival, or treatment-outcome data** exist.
6. **No Orphanet/ICD/MeSH codes** currently cross-referenced; nosology still consolidating.
7. **gnomAD/UniProt/MONDO claims** are database-derived and not independently re-validated here.

Proposed Follow-up Experiments / Actions

1. **Generate a genetic *Uba2*^+/− mouse** (and conditional neural-crest/limb-bud knockouts) to confirm haploinsufficiency recapitulates ACCES and to map tissue-specific requirements.
2. **Quantify global and substrate-specific SUMOylation** in patient-derived fibroblasts/iPSCs to directly test step 3 of the causal chain.
3. **Patient iPSC → neural-crest and limb-bud/organoid differentiation** to test the predicted neural-crest-repression and FGFR-dysregulation branches in a human system.
4. **Establish an international registry** to define prevalence, penetrance, genotype–phenotype correlations, natural history, and QoL.
5. **Systematic ClinVar/variant functional assays** (zebrafish rescue, in vitro E1 activity) to classify new *UBA2* VUS and clarify the SHFM-vs-ACCES determinants.
6. **Pharmacovigilance** for SUMO-E1 inhibitors (e.g., TAK981/subasumstat) regarding embryo-fetal toxicity, given the teratogenic mouse data.
7. **Pursue formal Orphanet/ICD-11 coding** to improve ascertainment and knowledge-base interoperability.

Evidence source legend: Human clinical (case reports/series, exome/CMA); Model organism (zebrafish null, mouse pharmacological); Database/computational (gnomAD constraint, UniProt biochemistry, MONDO/OLS4). This report integrates 9 confirmed findings and 20 reviewed papers across 5 investigation iterations.
